# BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SWAT

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2023

# MATHEMATICS

Past Paper 2023 • Paper Code: 102301

Retyped faithfully from the original board paper — same questions, no changes.

## SECTION-A

**Time Allowed:** 20 Minutes | **Maximum Marks:** 15

Note: 1) Attempting all MCQs is compulsory. This paper along with the OMR sheet must be returned to the superintendent after due time. 2) Fill the circle A B C D, which one is correct with blue or black ball point in separate OMR Sheet like ● 3) If more than one circle in the OMR sheet is filled then no credit will be given to such answer.

Q1. $1 + \tan^2\theta =$ _____.

(A) $\sin^2\theta$ (B) $\cos^2\theta$

(C) $\text{Cosec}^2\theta$ (D) $\sec^2\theta$

Q2. Phythagoras was a _____ Mathematician.

(A) Indian (B) Greek

(C) Pakistani (D) Chinese

Q3. A chord is 5 cm from the centre of a circle of radius 13 cm then length of chord is _____.

(A) 6 cm (B) 12 cm

(C) 24 cm (D) 30 cm

Q4. Two tangents drawn to a circle from a point outside it are of _____ in length.

(A) Half (B) Equal

(C) Double (D) Triple

Q5. How many Tangents can be drawn from the point outside the circle _____.

(A) 1 (B) 2

(C) 3 (D) 4

Q6. The most frequent value in the data is called _____.

(A) Mean
(B) Median
(C) Mode
(D) G.Mean

Q7. The reciprocal of Arithmetic Mean is _____.

(A) G. Mean
(B) Harmonic Mean
(C) Median
(D) Mode

Q8. If A has two elements and B has three elements, then number of binary relation in A x B is _____.

(A) 2 x 3
(B) $2^3$
(C) $2^6$
(D) $2^2$

Q9. Direct variation between a and b is expressed as _____.

(A) a = b
(B) a = 1/b
(C) a ∝ b
(D) a ∝ 1/b

Q10. $(2x^2+1)/(x-1)$ is _____ fraction.

(A) Improper rational
(B) Proper rational
(C) Irrational
(D) Proper irrational

Q11. The third proportional of x and y is _____.

(A) xy
(B) x/y
(C) $y^2/x$
(D) None of these

Q12. Discriminant of $4x^2 + x + 1 = 0$ is _____.

(A) 15
(B) -15
(C) 16
(D) 17

Q13. The sum of the cube roots of unity is _____.

(A) 1
(B) 2
(C) Zero
(D) ± 1

Q14. If $x^2 - 1/4 = 0$ Then solution are _____.

(A) X= ± 1/2
(B) x = ± 1/4
(C) x = ± 2/8
(D) x = ± 1/16

Q15. If $2^x = 1/2$ then X = _____.

(A) -1
(B) 0
(C) 2
(D) None of these

## SECTION "B"

**Time Allowed:** 2 hours and 40 minutes | **Maximum Marks:** 36

QII. Attempt any NINE Parts out of the following. Each Part carries equal marks.

(i) Solve for x. $(x-1)/(x+3) + (x+3)/(x-1) = 13/6$.

(ii) Solve for x. $\sqrt{x} + \sqrt{(3x+1)} = \sqrt{(5x+1)}$.

(iii) Find the value of K. If the sum of roots of $2x^2 + Kx + 6$ is equal to the product of its roots.

(iv) Show that $x^3 - y^3 = (x - y)(x - \omega y)(x - \omega^2 y)$.

(v) There is direct variation between $x^2$ and y. When x = 7, y = 49 find y when x = 9.

(vi) Find the Mean proportional of 12, 3.

(vii) Resolve into Partial fractions $1/(x(x^2+1))$.

(viii) If A = {1,2,3,-------12} and B = {2,4,6,-------12} verify commutative property of union and intersection.

(ix) Domain of a binary relation R= {(x,y) | y = 2x} is the set of {0,4,8} .Find Range of R.

(x) Find the standard Deviation for the following Data 6,8,10,12,14 .

(xi) Find l when $\theta = \pi/6$ radian , r = 2 cm.

(xii) Prove that $(\sec^2\theta - 1)\cos^2\theta = \sin^2\theta$.

## SECTION "C"

**Maximum Marks:** 24

> Note: Attempt any THREE questions of the following. Each question carries equal Marks.

QIII. Perpendicular from the centre of a circle on a chord bisect it.

QIV. Prove that, two tangents drawn from a point outside a circle are equal in length.

QV. Prove that, The angle in a Semi circle is a right angle.

QVI. Construct a Triangle with sides 2 cm, 2.5 cm and 3 cm. Also draw its Circum circle.